Drug Resistance Mutations Affect on HLA Binding in HIV-1 Epitopes

Harrison Deng

April 20, 2024

Abstract

This study makes use of the Human Immunodeficiency Virus Type 1 epitope dataset from Los Alamos HIV databases [15] and the drug resistant mutation scores from Stanford HIV Drug Resistance Mutation Database [21] to interrogate the intersection between such DRMs, and the human leukocyte antigen class I mediated, epitope suppressing immune escape mechanism. To elaborate, this study, on top of the naturally occurring epitopes, generates all possible permutations of the epitopes by introducing the previously mentioned drug resistant mutations when possible. The generated set of epitopes is ensured to be unique and run through NetMHCPan [20] against several common human leukocyte antigen class I alleles to predict the immunogenicity of all epitopes. Then, drug resistant mutation scores are calculated for each of the epitopes. The drug resistance and immunogenicity values are then categorized into different levels of drug resistance and binding respectively. Associative statistics on the number of times a specific set of mutations occur, and the resulting human leukocyte antigen class I binding levels are calculated amongst other forms of analysis. Our results indicate amongst integrase strand transfer inhibitor drugs, several drug resistant mutations results in strong binding towards HLA-B*15:01, and HLA-B*07:02. Similarly, for the protease inhibitor drug class, several drug resistant mutations were also determined to have an increased binding towards HLA-B*07:02 and thus, introduce a fitness burden to Human Immunodeficiency Virus Type 1 when selecting for such mutations.

Contents

1 Introduction & Background **3**
1.1 Drug Resistance 3
1.2 Immune Escape 3
1.3 Motivation for Study 4

2 Materials & Methods **5**
2.1 Materials 5
2.2 Methods 5

3 Results **8**
3.1 Comparing Distribution of Binding Strength 8
3.2 Binding of Original Epitopes vs. Generated Epitopes 9
3.3 Mutation Sets Associated with change in Binding 9

4 Discussion **13**
4.1 No Link Between Drug Resistance & Immunogenicity 13
4.2 Potential Increased Immunogenicity with HLA-B*15:01 13
4.3 Specific Drug Resistance Mutations Affects on Immunogenicity 14

5 Limitations **15**

6 Acronyms **16**

7 References **17**

1 Introduction & Background

HIV-1, a virus transmitted by body fluids, capable of shutting down the host immune system if not treated immediately [24]. As of 2020, an estimated 37 million people around the globe suffer from HIV-1 [30]. Furthermore, HIV-1 continues to spread, where it is estimated an additional 1.7 million global citizens acquire HIV-1 on average each year [30]. Despite having been discovered over 4 decades ago in 1981 [26], HIV-1, the cause of Acquired Immune Deficiency Syndrome (AIDS), remains an incurable disease and evades any form of vaccination or cure [30] where the best treatment option for patients of HIV-1 is to undergo antiretroviral therapy (ART) treatment [18]. With ART, HIV-1 associated mortality rates have been successfully reduced ultimately saving millions, and changing AIDS from a deadly disease to one where the patients may typically live a full life [30]. However, it is critical to accept ART not as a complete solution, but rather, as a temporary solution. ART itself remains an imperfect treatment with problems ranging from rebound due to patients not fulfilling the required periodic treatment plan [2], to constrained availability of treatment [23], to problematic cost of treatment [11, 26].

1.1 Drug Resistance

Amongst these problems, however, there underlies a biologically critical change HIV-1 is undergoing common to many pathogens where treatment is not potent enough to cure is the gold standard and thus deterring the progress of ART. Specifically, in recent years, with the advent of ART, HIV-1, has been slowly but surely developing resistance to many modern ART treatment-related drugs [3, 5, 8].

To elaborate on HIV-1 DRMs, it is typical that a single amino acid variation coded for in the HIV-1 genome is not enough to induce resistance towards antiretroviral drugs (ARDs) [3]. Furthermore, within a given drug classification, it is rather common to see cross-resistance where the increased resistance towards one particular drug increases the resistance towards another [3]. Despite such cross-resistance prevalence within the same drug class, the same cannot be said across different drug classes where current research seems to indicate that it is more difficult [12] for such cross-resistances, however, not impossible [3, 14, 25]. This lower prevalence of cross-class drug resistance is a key reason for the use of multiple ARDs of differing classes during modern ART treatment plans [7]. Notably, such DRMs seem to typically result in a fitness penalty to the host strain of the DRMs [3].

1.2 Immune Escape

HIV-1 is fascinating in that it is one of the few viruses in the modern day that both is highly prevalent in the population, but also eludes cures. Specifically, Cytotoxic T Cell (CTL), also known as CD8+ T cells, are well known to be a host's main antagonist to theHIV-1 virus [6]. Then, it goes to reason, that HIV-1, with a relatively unstable genome capable of high rates of proliferation [22], since its initial entry into the human population, has developed various mechanisms of circumvent, or escape, an effective CTL response. One such mechanism is via certain specific immune escape mutations (IEMs) that result in the suppression of displaying specifically HIV-1 associated antigens thus avoiding apoptosis via natural killer T cells [29]. Similarly to DRMs, IEMs also seem to introduce a fitness penalty to the HIV-1 strain [13, 33].

1.3 Motivation for Study

Both mutations, independent of each other are relatively thoroughly studied. However, at the time of writing, there exists a minimal amount of studies looking into the intersection between both DRMs and IEMs. In other words, since both forms of mutations incur fitness penalties, an area of interest may be how DRMs affects the mutation-based immune escape process via epitopic modifications. The few studies emphasizing such a topic discuss various DRMs that both corroborate the increase of immunogenicity of HIV-1 [1, 16], and a decrease in immunogenicity, depending on varying factors [17, 19]. The results of these studies are reasonable as DRMs occurring within epitope regions, i.e., regions that affect the presentation via HLA-I may modulate the fitness penalty due to directly affecting an immune escape mechanism. This study hopes to extend the knowledge base surrounding such interplay between DRMs and IEMs as such knowledge may serve as foundational to improving treatment regimens by selecting drugs that produce strong fitness penalties and the future development of HIV-1 vaccines, as explained by previous studies [17, 19].

2 Materials & Methods

2.1 Materials

Analysis of the interplay between DRMs, and IEMs were performed upon datasets upon the LANL-HIV and the DRM-DB. NetMHCPan version 4.1 and an in-house Python script were used in combination for accommodating large data throughput [20]. Within the Python script, SciPy and Pandas data processing libraries were used for the analysis [28, 31]. Figures were generated with the use of Matplotlib and Seaborn [9, 32].

The LANL-HIV dataset of known HIV-1 CTL epitopes including the peptide sequence and position relative to the gene. Datasets were individually downloaded by querying for integrase, p24, protease, and Reverse Transcriptase (RT) while all other search parameters were left as the most broadcasting option i.e., selecting the "ALL" option when present. The DRM-DB amongst other tools, and other datasets, contains a scoring mechanism for known drug resistance-inducing mutations in the HIV-1 genome [21]. Spreadsheets of such scoring for the different drugs concerning a given single mutation were downloaded. NetMHCPan version 4.1 was used to predict binding against different HLA-I alleles [20]. Data ingress and egress, manipulation and vectorized operations were performed by Pandas and the SciPy library was used for statistical associations [28, 31]. Lastly, Seaborn was used for the automated generation and regression of figures by making use of Matplotlib [9, 32].

2.2 Methods

All aforementioned datasets were imported from respective comma separated values (CSV) sheets downloaded from their respective databases. This was then followed by the generation of new epitopes based on the set of known epitopes, and the set of drug-resistant mutations. All epitopes, natural or generated were then run through NetMHCPan 4.1 to predict their binding level towards various HLA-I alleles. Such binding levels were fit into categories. Hypothetical epitopes were generated from the N-EPIs. All generated epitopes were cleaned and filtered to increase the number of interpretable natural epitopes and reduce computation time respectively before running the dataset in NetMHCPan [20]. DRMs and their associated resistance scores were summed in accordance with the guidelines of the DRM-DB. The resistance scores and DRM scores were then categorized according to their respective guidelines from DRM-DB. Association between the presence of DRMs and the binding level with various HLA-I alleles were tested and p-values were aggregated.

The first step involved downloading the data and parsing the data into Python. Data was procured from both LANL-HIV and DRM-DB as described in the materials section. DRM were further parsed and merged into a large, master spreadsheet where the drugs irrelevant to a mutation were set to have a DRM of 0. Similarly, known N-EPIs were also parsed. Unused columns of metadata were dropped and epitopes with the exact identical sequences were dropped immediately to ensure that any associations found would be purely based on their occurrence in an epitope, and not biased by the frequency in a population. Furthermore, the LANL-HIV contained synthetic epitopes created by concatenating two epitopes together. Such epitopes did not have well-defined bounds and thus, also did not have well-defined DRMs [15]. Such junction-based epitopes had their junctions removed and the original epitope restored by selecting the portion of the junction epitope based on the annotated bound.

The unique set of naturally occurring HIV epitopes then had variants generated naively simply by generating all possible different combinations of mutations that could be added to a naturally occurring epitope. These epitopes were then added to a new flat database with their original DRMs, i.e., mutations that existed before any modification, and the added mutations. Notably,

due to some epitopes having many possible combinations of DRMs, constraints were added such that generated epitopes that resulted in a DRM score lower than 15 were removed. This corresponds to DRM-DBs categorization of having none, or potentially a very low level of drug resistance (Table 2). Additionally, some N-EPI had many peptide positions with known DRMs and thus, a ranked threshold was applied where the DRM scores were used to rank the generated epitopes and such epitopes that ranked higher than 1000 were removed. I.e., the top 1000 highest DRM scoring generated epitopes were kept.

Then, the produced aggregate of both N-EPIs and the generated epitopes had binding levels with all HLA-I alleles that were recognized by NetMHCPan 4.1 (table 1) predicted and cataloged [20]. The resulting database had DRM scores and eluted ligand percentile rank (%EL-Rank) [10] calculated and mapped according to table 2 and to table 3 respectively (for more information on this metric, see original software paper for version 4.0) [10, 20]. Furthermore, changes in binding level were calculated by comparing the N-EPIs with the generated ones.

Default HLA Variants
HLA-A02:01
HLA-A01:01
HLA-A03:01
HLA-A24:02
HLA-A26:01
HLA-B07:02
HLA-B08:01
HLA-B27:05
HLA-B39:01
HLA-B40:01
HLA-B58:01
HLA-B*15:01

Table 1: The Default list of HLA alleles for which NetMHCPan is capable of predicting %EL-Rank [20].

Resistance Category	DRM Score (Inclusive)
No drug resistance	0 to 9
Potential low-level drug resistance	10 to 14
Low-level drug resistance	15 to 29
Intermediate resistance	30 to 59
High-level resistance	60 and above

Table 2: The categorical levels of drug resistance based on the DRM score assigned for an entire sequence as defined by DRM-DB.

.

Lastly, all combinations between all sets of DRMs, and the empty set of DRMs, i.e., the lack of DRMs, were tested for association with all different levels of binding, that is to say, these combinations of DRMs were tested against the occurrences of weak, strong, and no binding level epitopes. Fisher's exact test was used due to the relatively small sample count of epitopes with the same set of DRMs with unique epitopes. This process generated a list of different p-values for each possible combination. This list of p-values was then filtered by a $\alpha = 0.05$.

Resistance Category	DRM Score (Inclusive)
Strong Binding	Less than 0.5
Weak Binding	Less than 2, greater than or equal to 0.5
No Binding	Other

Table 3: The categorical levels of HLA binding based on the percentile rank predicted by NetMHC-Pan 4.1 where both the metric and the thresholds were advised by the original paper and authors of the software [20].

Resulting tables were generated via Pandas and resulting figures were generated with Seaborn and Matplotlib [9, 28, 32]. Further post-analysis filtering and selection of datasets were also performed by these libraries.

3 Results

Overall, a total of 390,816 unique epitopes were used for the following interpretation and analysis. From these 390,816 epitopes, 202,740 unique sets of epitopic DRMs were generated.

3.1 Comparing Distribution of Binding Strength

Figure 1 is a histogram where the vertical axis is the proportion of all DRM generated epitopes binned based on %EL-Rank with 50 bins resulting in every 2 percentiles being encapsulated in one bin. The N-EPI and their %EL-Rank are overlaid in orange with identical binning parameters. See that by table 3, the first bin contains all binding values. Such a figure indicates over all DRMs generated unique epitopes, there is a shift of a portion of epitopes towards being non-binding towards HLA-I. Furthermore, see the substantial drop in binding epitopes in contrast with the N-EPI.

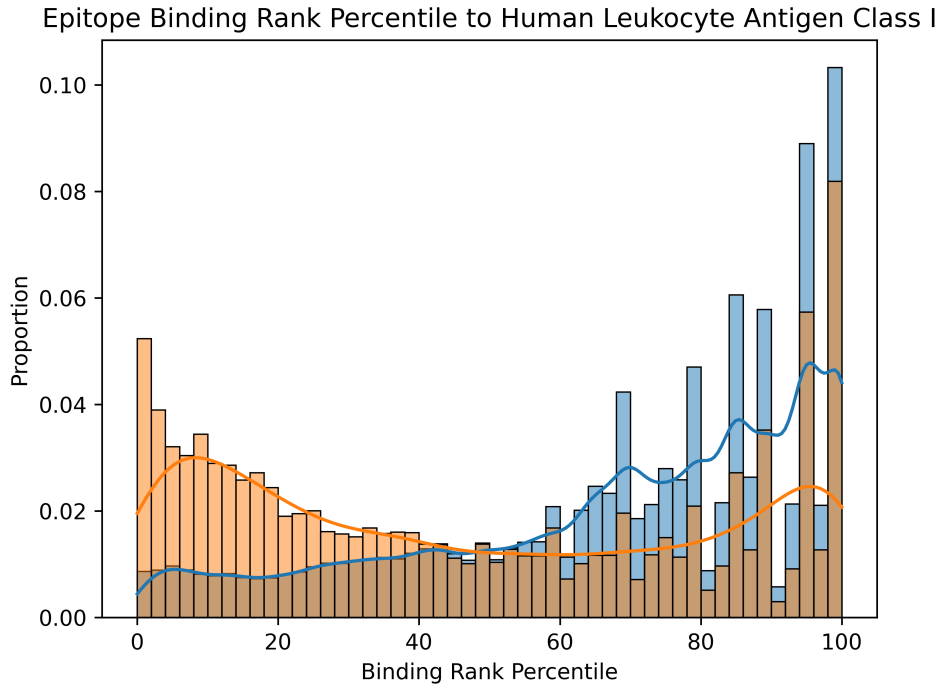


Figure 1: Histogram of generated epitopes based on drug resistant mutations binned by eluted ligand percentile rank. The orange is the distribution of the naturally occurring epitope binned based on their binding strengths while the blue represents the distribution of the generated epitopes.

However, filtering out results to include only generated epitopes with a maximum of 2 DRMs, the maximum number of DRMs that occurred within the set of N-EPIs for any given N-EPIs, the histogram becomes figure 2. Figure 2, in contrast with figure 1, produces a histogram where the distribution of DRM generated epitopes is much more similar to the distribution of the N-EPI.

Note in both figures 1 and 2, that there is proportionally more N-EPIs binding to HLA-I in contrast with the DRM generated set of epitopes.

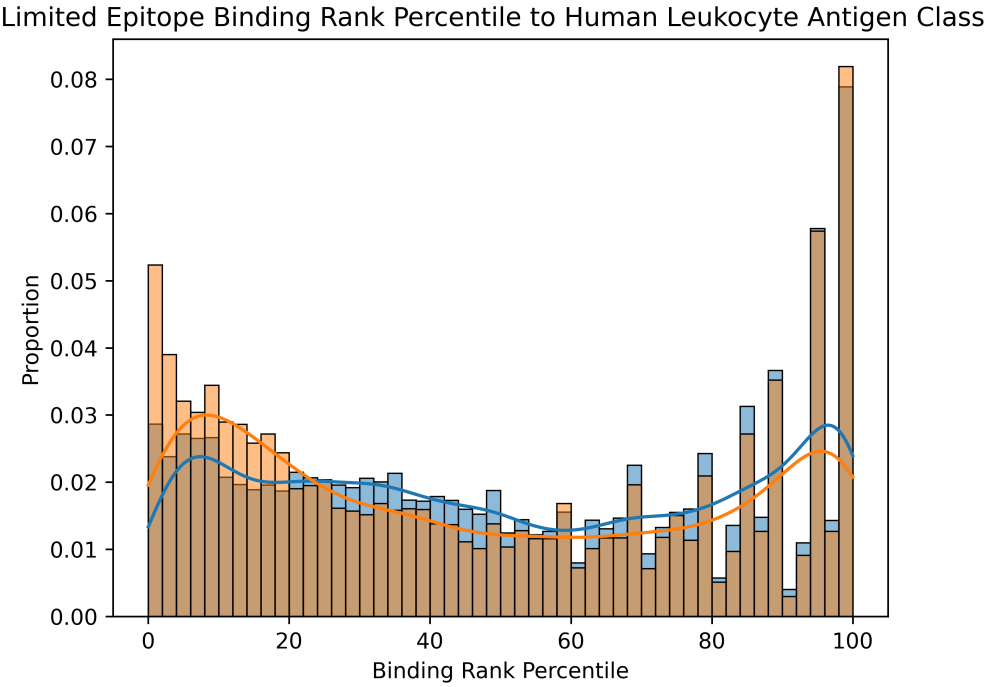


Figure 2: Histogram of generated epitopes with at most 2 drug resistant mutations with 50 bins by eluted ligand percentile rank. The orange is the distribution of the naturally occurring epitope binned based on their binding strengths while the blue represents the distribution of the generated epitopes.

3.2 Binding of Original Epitopes vs. Generated Epitopes

Despite the lack of any obvious relation between DRMs, when the data is stratified by drug class, and then stratified further by the HLA-I allele, we can see that some HLA-I variants for some drug classes do show a non 1:1 association between the DRM generated epitope, and the corresponding N-EPI. This can be seen in figure 3 where some HLA-I alleles do not grow in a 1:1 fashion. Namely, aside from the kernel density estimate (KDE) plots that mostly seem to be a line with a slope of one, there are quite a few where the mutated version has a greater distribution in the direction of higher %EL-Rank when compared to the original. However, there are also a few groupings that result in the distribution shifting towards a lower rank, and thus, higher binding strength.

3.3 Mutation Sets Associated with change in Binding

With regards to statistics and mutation set association results, we will look at the top 13 most significant results that contain dramatic binding changes, i.e., binding changes that go from no binding to strong binding, or vice versa. Table 4 indicates the counts of sets of mutations occurring in relation to different binding strengths. Of note, see that of the top 13 most significant associations with dramatic changes in binding level, 11 results indicate a form of binding with HLA-I variant HLA-I-B07:02. Furthermore, of these results, all indicate that the insertion of such mutations result in an increase in binding where previously, there may have not been any form of binding, these mutations introduce strong binding with HLA-I.

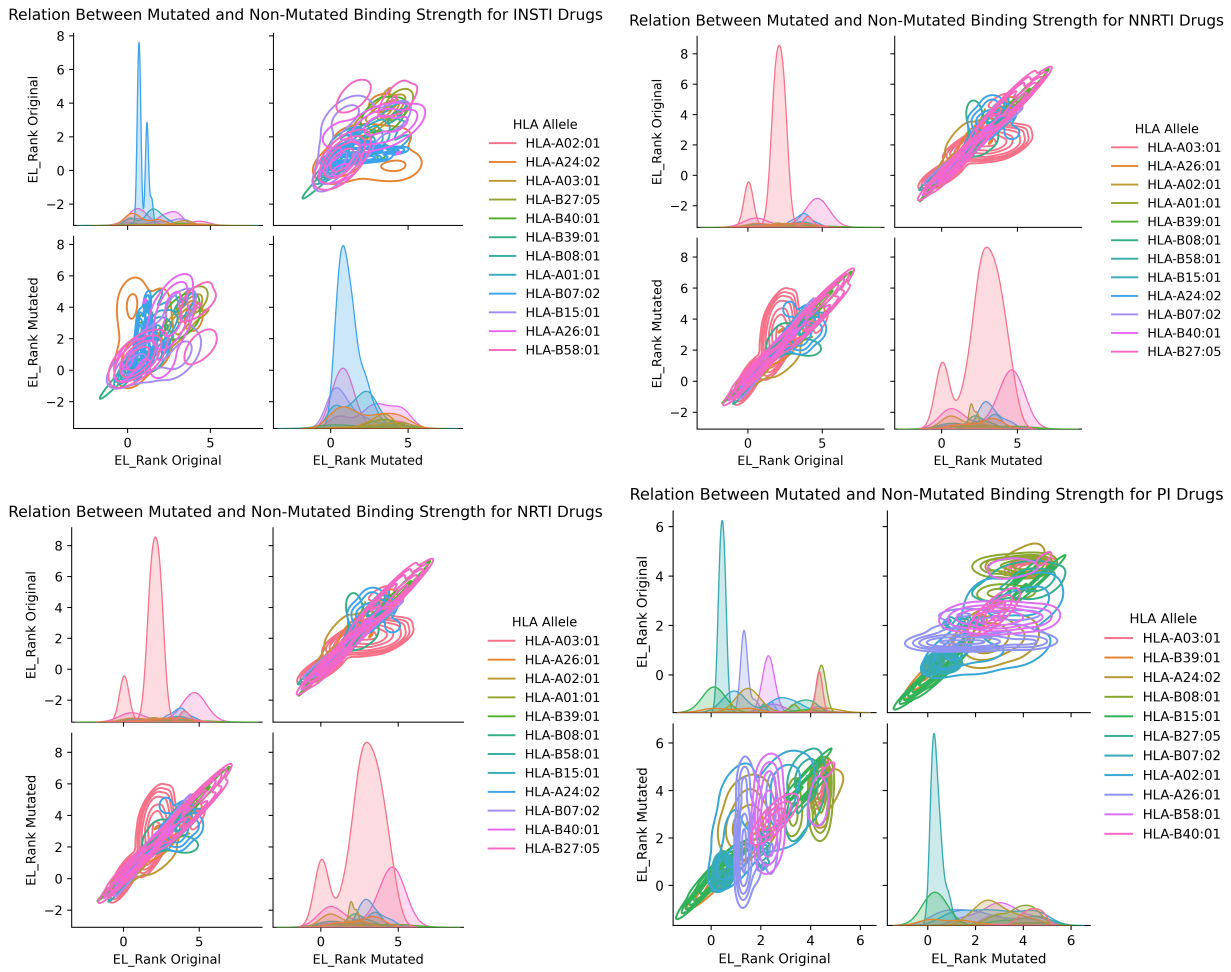


Figure 3: kernel density estimate plots of the relation between the change in binding affinity towards various human leukocyte antigen class I alleles. Only epitopes with at most 2 drug resistant mutations, had an original binding rank of less than or equal to 5 and had a mutated binding rank of less than or equal to 5 were included in the plotting. Assume all alleles have the asterisk (*) separator.

Furthermore, see that for each result mutation set (table 5), the score of these sets cumulatively, has at least one drug of the respective class that is considered strongly drug resistant.

HLA Allele	HLA-B*07:02	**2-Way Table**	**SB**	**NB**
Drug Target	Integrase	α **&** β*****	2	0
P-Value	0.00026	**No DRMs**	2	209
HLA Allele	HLA-B*07:02	**2-Way Table**	**SB**	**NB**
Drug Target	Integrase	α **&** β†	2	1
P-Value	0.00078	**No DRMs**	2	209
HLA Allele	HLA-B*15:01	**2-Way Table**	**SB**	**NB**
Drug Target	Integrase	**E92Q & Q95K**	3	8
P-Value	0.00080	**No DRMs**	2	215
HLA Allele	HLA-B*15:01	**2-Way Table**	**SB**	**NB**
Drug Target	Integrase	**E92Q**	3	9
P-Value	0.00104	**No DRMs**	2	215
HLA Allele	HLA-B*07:02	**2-Way Table**	**SB**	**NB**
Drug Target	Protease	**N88T & V82M**	2	0
P-Value	0.00111	**No DRMs**	1	71

*****: α and β may be of any of the following combinations (6): (Q**146**P, Q**148**R) (Q148H, Y143K), (Q148H, Y143R), (Q148K, Y143R), (Q148N, Y143R), (Q148R, Y143R)
†: α and β may be of any of the following combinations (4): (Q146P, Y143H), (Q146P, Y143K), (Q146P, Y143R), (S**147**G, Y143R)

Table 4: The top 13 most statistically significant associations between a given set of mutations existing in all unique variations of epitopes, with a change of eluted ligand percentile rank such that the categorical binding strength changes from strong binding (SB) to no binding (NB), or vice versa. Note that due to some of the associations having the same 2-way table, with the same properties other than the mutations, they were condensed into a list to save space. Peptide positions that are different from the rest are bolded solely for legibility.

Drug Target: Integrase

HLA	Mutation Set	Binding	BIC	CAB	DTG	EVG	RAL
B07:02	Q146P, Q148R	SB, NB	25	70	25	120	60
B07:02	Q148H, Y143K	SB, NB	30	40	30	70	120
B07:02	Q148H, Y143R	SB, NB	30	40	30	70	120
B07:02	Q148K, Y143R	SB, NB	35	60	35	70	120
B07:02	Q148N, Y143R	SB, NB	5	25	5	25	75
B07:02	Q148R, Y143R	SB, NB	30	50	30	70	120
B07:02	Q146P, Y143H	SB, NB	5	40	5	70	60
B07:02	Q146P, Y143K	SB, NB	5	40	5	70	60
B07:02	Q146P, Y143R	SB, NB	5	40	5	70	60
B07:02	Q146G, Y143R	SB, NB	15	20	15	70	70
B15:01	E92Q, Q95K	SB, NB	15	20	15	70	70
B15:01	E92Q	SB, NB	10	15	10	60	30

Drug Target: Protease

HLA	Mutation Set	Binding	ATV/r	DRV/r	LPV/r	FPV/r
B07:02	Q146P, Q148R	SB, NB	25	0	25	15

HLA	Mutation Set	Binding	IDV/r	NFV	SQV/r	TPV/r
B07:02	Q146P, Q148R	SB, NB	30	60	15	10

Table 5: Drug resistance scores according to the Stanford HIV Drug Resistance Mutation Database corresponding to the top 13 most significant results. Both mutation set column and binding columns correspond to their respective information in table 4. Where SB indicates strong binding, and NB indicates a lack of binding. Note that the Protease table has been broken into two lines.

4 Discussion

4.1 No Link Between Drug Resistance & Immunogenicity

Currently, It is well established that one mechanism for which HIV-1 evades the immune system are mutations that inhibit the body's natural ability to display antigenic peptides via HLA-I [29]. Furthermore, it is also well established that HIV-1 is, more and more, developing mutations capable of reducing the effectiveness of a variety of drugs [3]. Lastly,

It then may be interesting to see if there does exist a general trend of any sort with regards to DRMs in HIV-1 epitope regions. From figure 1, we see that the distribution over %EL-Rank has proportionately, shifted towards higher rankings when compared to the N-EPIs, and thus a lower likelihood of the epitopes successfully binding with HLA-I resulting in the suppression of viral peptide presentation, while not disturbing the functionality of the HLA-I in its natural state. However, these results assume that all hypothetical epitopes are virally viable. In this scenario, HIV-1 would not only be resistant towards the majority of known ARD, but also be very accomplished in evading the host's immune response thus producing monster HIV-1. However, it has been established that both forms of mutation, either to escape an effective immune response or to resist the effects of ARDs, will typically result in some form of penalty to the fitness of the HIV-1 strand [3, 13, 33]. From this then, for a host undergoing ART treatment, the HIV-1 may be said to be undergoing selective pressures from the host immune system, the drug, and its own functional viability, that is to say, the incurring of too many mutations may be deleterious to the innate functionality of proteins encoded for by the strand of HIV-1 itself. It is this last broad selective pressure that is likely preventing such a monster HIV-1 from actually forming. In other words, the fitness burden of maintaining such a degree of DRMs within the HIV-1 genome must be deleterious to its functionality.

Thus, a more realistic distribution would involve filtering the generated epitope set solely for generated epitopes with the same, or less DRMs than that of the greatest number of DRMs in the set of N-EPIs. By doing this, are essentially naively assuming for every DRM, the DRM can be said to carry 1 fitness burden unit. Then, by resolving for the maximum number of DRMs in the set of N-EPIs, we essentially attempt to find the upper viable limit of DRM based fitness burden units per epitope where going over is less optimal then being more susceptible due to not having a given DRM. The resulting figure 2 indicates that in the case that such an assumption holds relatively true, the overall effect of adding at most two DRMs to any given epitope does not affect greatly affect the population of epitope's capacity to bind or rather, avoid the activation of HLA-I, and thus, does not affect HIV-1's immunological escape capacities via the suppressing of peptide presentation via HLA-I.

4.2 Potential Increased Immunogenicity with HLA-B*15:01

Despite the overall lack of association between the presence of at most 2 DRMs in epitopes, by stratifying the data and representing the data in the form of a KDE, we can firstly, see the initial conclusion of the overall distribution of %EL-Rank not greatly affecting remains visible in each subplot of figure 3. I.e., the figure plots the %EL-Rank of all generated epitopes and the %EL-Rank of the N-EPI the generated epitope originated from (root epitope). This, when not accounting for the stratification of different HLA-I alleles, we see that as the rank of root epitopes increases over the horizontal axis, the rank of the mutated epitopes grows at a similar rate with some deviation. However, stratifications that result in distributions that do not follow this trend also exist. Notably, there does seem to exist some sets of root epitopes where the introduction of DRMs resulted in a decrease of %EL-Rank thus increasing binding strength to the respective allele. Specifically, of the

INSTI class, we see this in the HLA-B*15:01 allele. In contrast, and potentially as expected, the majority of the root epitopes, when introduced with DRMs, result in an increase in the spread of distribution towards an increased %EL-Rank, thus, indicating that many DRMs may up-regulate immune escape mechanisms assuming that the combination of DRMs are viable.

HLA-B*15:01 is not prevalently discussed with regards to HIV-1. Interestingly, one paper has previously associated HLA-B*15:01 with an increased susceptibility to HIV-1 in a Turkish population [4]. However, this study is not regarding HIV-1 susceptibility, but rather, the effect of DRMs on the immunogenicity of HIV-1. In other words, HIV-1, in acquiring DRMs to counteract INSTI drugs, may be more susceptible to host HLA-B*15:01 host immune response. Alternatively, if it is true that HLA-B*15:01 increases susceptibility to HIV-1 patients, it may also be true that INSTI drugs will be more effective for those patients in that INSTI drugs may place selective pressure on the HIV-1 strand to develop drug resistance towards INSTI, however, in doing so, may increase such a hypothetical strand's HIV-1 susceptibility to HLA-B*15:01 based host immune response.

4.3 Specific Drug Resistance Mutations Affects on Immunogenicity

To begin, we will look at the effects of HLA-B*07:02 on INSTI class ARDs. Notably, multiple papers discuss the protective properties of HLA-B*07:02 towards HIV-1 [4, 27]. Our results indicate several combinations of DRMs that may be associated with increased immunogenicity by increasing the %EL-Rank, and thus undermining potential previously HLA-I mediated immune escapes. The association with a strong increase in immunogenicity is corroborated by the DRM-DB where it indicates such a mutation is rather rare. This rareness may be justified due to its high penalty to HIV-1's immune escape capabilities. Furthermore, all mutations within this general region are categorized as highly resistant to most INSTI drugs (table 5 for resistance scores and table 2 for interpreting the scores).

Continuing, HLA-B*15:01 appears again where specifically, mutations set E92Q and Q95K, appear together, and E92Q itself, is associated with a relatively dramatic increase in immunogenicity with high drug resistance capabilities. A separate paper has replicated this result, concluding similarly, however, for a specific peptide [19]. Furthermore, of note, HLA-B*15:01, which was discussed previously as part of the analysis of the joint KDE plot, supports each other in that both seem to indicate an increase in HLA-I %EL-Rank decrease, and thus, a binding strength increase, ultimately, predicting an increase of immunogenicity.

Lastly, N88T and V82M protease epitope mutations were found to be statistically significantly associated with an increase in predicted immunogenicity towards the HLA-B*07:02 HLA-I. Interestingly, DRM-DB notes that N88T is considered extremely rare while only conferring minor resistance towards select PI drugs. Similarly, V82M is considered uncommon by DRM-DB. The rareness corroborates our results as we indicate, as a result of the two mutations, a strong increase in immunogenicity. Furthermore, the combination of the two only significantly confers strong resistance towards Nelfinavir according to DRM-DB.

5 Limitations

This study was performed entirely in silico with data produced from other sources. As such, these results intend to serve as a guide for future studies in terms of what to look at and where to look in terms of ARD that may place selective pressure on HIV-1 to result in compromises for the benefit of the patient. Furthermore, the statistics performed do not use a complete sample of all possible permutations of epitopes favoring strong DRMs scores. Introducing a filter for the maximum number of DRM mutations may accommodate this slightly, but future tests would hope to either perform the entire sample set or perform random sampling.

6 Acronyms

%EL-Rank eluted ligand percentile rank

AIDS Acquired Immune Deficiency Syndrome

ARD antiretroviral drug

ART antiretroviral therapy

CSV comma separated values

CTL Cytotoxic T Cell

DRM drug resistant mutation

DRM-DB Stanford HIV Drug Resistance Mutation Database

HIV-1 Human Immunodeficiency Virus Type 1

HLA-I human leukocyte antigen class I

IEM immune escape mutation

INSTI integrase strand transfer inhibitor

KDE kernel density estimate

LANL-HIV Los Alamos HIV databases

N-EPI naturally occurring epitope

PI protease inhibitor

RT Reverse Transcriptase

7 References

[1] Juan Blanco-Heredia et al. "Identification of Immunogenic Cytotoxic T Lymphocyte Epitopes Containing Drug Resistance Mutations in Antiretroviral Treatment-Naïve HIV-Infected Individuals". In: *PloS One* 11.1 (2016), e0147571. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0147571. pmid: 26808823.

[2] Kevin G. Buell et al. "Lifelong Antiretroviral Therapy or HIV Cure: The Benefits for the Individual Patient". In: *AIDS Care* 28.2 (Feb. 1, 2016), pp. 242–246. ISSN: 0954-0121. DOI: 10.1080/09540121.2015.1074653. pmid: 26357912. URL: https://doi.org/10.1080/09540121.2015.1074653 (visited on 04/19/2024).

[3] Dana S. Clutter et al. "HIV-1 Drug Resistance and Resistance Testing". In: *Infection, Genetics and Evolution: Journal of Molecular Epidemiology and Evolutionary Genetics in Infectious Diseases* 46 (Dec. 2016), pp. 292–307. ISSN: 1567-7257. DOI: 10.1016/j.meegid.2016.08.031. pmid: 27587334.

[4] Sule Darbas et al. "Relationship of HLA-B Alleles on Susceptibility to and Protection from HIV Infection in Turkish Population". In: *Northern Clinics of Istanbul* 10.1 (Dec. 13, 2022), pp. 67–73. ISSN: 2148-4902. DOI: 10.14744/nci.2021.00018. pmid: 36910436. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9996654/ (visited on 04/20/2024).

[5] Monique R. Ferguson et al. "HIV-1 Replication Cycle". In: *Clinics in Laboratory Medicine* 22.3 (Sept. 2002), pp. 611–635. ISSN: 0272-2712. DOI: 10.1016/s0272-2712(02)00015-x. pmid: 12244589.

[6] Victor Garcia and Roland Robert Regoes. "The Effect of Interference on the CD8+ T Cell Escape Rates in HIV". In: *Frontiers in Immunology* 5 (2015). ISSN: 1664-3224. URL: https://www.frontiersin.org/articles/10.3389/fimmu.2014.00661 (visited on 10/18/2023).

[7] Kevin M. Gibas et al. "Two-Drug Regimens for HIV Treatment". In: *The lancet. HIV* 9.12 (Dec. 2022), e868–e883. ISSN: 2352-3018. DOI: 10.1016/S2352-3018(22)00249-1. pmid: 36309038.

[8] "HIV-1 Genetic Variation and Drug Resistance Development". In: *Expert Review of Anti-infective Therapy* 11.11 (2013), pp. 1159–1178. ISSN: 1478-7210. DOI: 10.1586/14787210.2013.844649. URL: https://journals.scholarsportal.info/details/14787210/v11i0011/1159_hgvadrd.xml (visited on 10/18/2023).

[9] J. D. Hunter. "Matplotlib: A 2D Graphics Environment". In: *Computing in Science & Engineering* 9.3 (2007), pp. 90–95. DOI: 10.1109/MCSE.2007.55.

[10] Vanessa Jurtz et al. "NetMHCpan 4.0: Improved Peptide-MHC Class I Interaction Predictions Integrating Eluted Ligand and Peptide Binding Affinity Data". In: *Journal of immunology (Baltimore, Md. : 1950)* 199.9 (Nov. 1, 2017), pp. 3360–3368. ISSN: 0022-1767. DOI: 10.4049/jimmunol.1700893. pmid: 28978689. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5679736/ (visited on 04/15/2024).

[11] James G Kahn et al. "Cost-Effectiveness of Antiretroviral Therapy for Prevention". In: *Current HIV Research* 9.6 (Sept. 2011), pp. 405–415. ISSN: 1570-162X. DOI: 10.2174/157016211798038542. pmid: 21999776. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3529401/ (visited on 06/19/2023).

[12] Brendan A. Larder. "Interactions Between Drug Resistance Mutations in Human Immunodeficiency Virus Type 1 Reverse Transcriptase". In: *Journal of General Virology* 75.5 (1994), pp. 951–957. ISSN: 1465-2099. DOI: 10.1099/0022-1317-75-5-951. URL: https://www.microbiologyresearch.org/content/journal/jgv/10.1099/0022-1317-75-5-951 (visited on 10/18/2023).

[13] A. J. Leslie et al. "HIV Evolution: CTL Escape Mutation and Reversion after Transmission". In: *Nature Medicine* 10.3 (3 Mar. 2004), pp. 282–289. ISSN: 1546-170X. DOI: 10.1038/nm992. URL: https://www.nature.com/articles/nm992 (visited on 10/18/2023).

[14] Francesca Lombardi et al. "Prevalence and Factors Associated with HIV-1 Multi-Drug Resistance over the Past Two Decades in the Italian ARCA Database". In: *International Journal of Antimicrobial Agents* 57.2 (Feb. 2021), p. 106252. ISSN: 1872-7913. DOI: 10.1016/j.ijantimicag.2020.106252. pmid: 33259914.

[15] Jennifer L Mamrosh et al. "HIV Molecular Immunology 2022". In: (2022).

[16] Rosemarie D. Mason et al. "Antiretroviral Drug Resistance Mutations Sustain or Enhance CTL Recognition of Common HIV-1 Pol Epitopes". In: *Journal of Immunology (Baltimore, Md.: 1950)* 172.11 (June 1, 2004), pp. 7212–7219. ISSN: 0022-1767. DOI: 10.4049/jimmunol.172.11.7212. pmid: 15153547.

[17] Sandra M. Mueller et al. "Influence of Major HIV-1 Protease Inhibitor Resistance Mutations on CTL Recognition". In: *Journal of Acquired Immune Deficiency Syndromes (1999)* 56.2 (Feb. 1, 2011), pp. 109–117. ISSN: 1944-7884. DOI: 10.1097/QAI.0b013e3181fe946e. pmid: 21107269.

[18] Nittaya Phanuphak and Roy M. Gulick. "HIV Treatment and Prevention 2019: Current Standards of Care". In: *Current Opinion in HIV and AIDS* 15.1 (Jan. 2020), p. 4. ISSN: 1746-630X. DOI: 10.1097/COH.0000000000000588. URL: https://journals.lww.com/co-hivandaids/Abstract/2020/01000/HIV_treatment_and_prevention_2019__current.3.aspx (visited on 06/19/2023).

[19] Mohammad Arif Rahman et al. "Raltegravir and Elvitegravir-Resistance Mutation E92Q Affects HLA-B*40:02-Restricted HIV-1-specific CTL Recognition". In: *Microbes and Infection* 16.5 (May 2014), pp. 434–438. ISSN: 1769-714X. DOI: 10.1016/j.micinf.2014.03.003. pmid: 24657622.

[20] Birkir Reynisson et al. "NetMHCpan-4.1 and NetMHCIIpan-4.0: Improved Predictions of MHC Antigen Presentation by Concurrent Motif Deconvolution and Integration of MS MHC Eluted Ligand Data". In: *Nucleic Acids Research* 48.W1 (July 2, 2020), W449–W454. ISSN: 0305-1048. DOI: 10.1093/nar/gkaa379. pmid: 32406916. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7319546/ (visited on 04/15/2024).

[21] Soo-Yon Rhee et al. "HIV-1 Transmitted Drug Resistance Surveillance: Shifting Trends in Study Design and Prevalence Estimates". In: *Journal of the International AIDS Society* 23.9 (Sept. 2020), e25611. ISSN: 1758-2652. DOI: 10.1002/jia2.25611. pmid: 32936523.

[22] Douglas D. Richman et al. "HIV Evolution and Escape". In: *Transactions of the American Clinical and Climatological Association* 115 (2004), pp. 289–303. ISSN: 0065-7778. pmid: 17060974.

[23] Erik J. Schouten et al. "Antiretroviral Drug Supply Challenges in the Era of Scaling up ART in Malawi". In: *Journal of the International AIDS Society* 14 Suppl 1 (July 6, 2011), S4. ISSN: 1758-2652. DOI: 10.1186/1758-2652-14-S1-S4. pmid: 21967844.

[24] F. Douglas Scutchfield and A. S. Benenson. "AIDS Update". In: *Postgraduate Medicine* 85.4 (Mar. 1, 1989), pp. 289–304. ISSN: 0032-5481. DOI: 10.1080/00325481.1989.11700636. URL: https://doi.org/10.1080/00325481.1989.11700636 (visited on 04/19/2024).

[25] Sorana Segal-Maurer et al. "Capsid Inhibition with Lenacapavir in Multidrug-Resistant HIV-1 Infection". In: *The New England Journal of Medicine* 386.19 (May 12, 2022), pp. 1793–1803. ISSN: 1533-4406. DOI: 10.1056/NEJMoa2115542. pmid: 35544387.

[26] Paul M. Sharp and Beatrice H. Hahn. "Origins of HIV and the AIDS Pandemic". In: *Cold Spring Harbor Perspectives in Medicine:* 1.1 (Sept. 2011), a006841. ISSN: 2157-1422. DOI: 10.1101/cshperspect.a006841. pmid: 22229120. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3234451/ (visited on 06/05/2023).

[27] Jagadish Chandrabose Sundaramurthi, V. D. Ramanathan, and Luke Elizabeth Hanna. "HLA-B*27:05-Specific Cytotoxic T Lymphocyte Epitopes in Indian HIV Type 1C". In: *AIDS research and human retroviruses* 29.1 (Jan. 2013), pp. 47–53. ISSN: 1931-8405. DOI: 10.1089/AID.2011.0374. pmid: 22924625.

[28] The pandas development team. *Pandas-Dev/Pandas: Pandas*. Version v2.2.2. Zenodo, Apr. 2024. DOI: 10.5281/zenodo.10957263. URL: https://doi.org/10.5281/zenodo.10957263.

[29] Angel Varela-Rohena et al. "Control of HIV-1 Immune Escape by CD8 T Cells Expressing Enhanced T-cell Receptor". In: *Nature Medicine* 14.12 (Dec. 2008), pp. 1390–1395. ISSN: 1078-8956, 1546-170X. DOI: 10.1038/nm.1779. URL: https://www.nature.com/articles/nm.1779 (visited on 10/18/2023).

[30] John D. Ventura. "Human Immunodeficiency Virus 1 (HIV-1): Viral Latency, the Reservoir, and the Cure". In: *The Yale Journal of Biology and Medicine* 93.4 (Sept. 30, 2020), pp. 549–560. ISSN: 0044-0086. pmid: 33005119. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7513431/ (visited on 06/04/2023).

[31] Pauli Virtanen et al. "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python". In: *Nature Methods* 17 (2020), pp. 261–272. DOI: 10.1038/s41592-019-0686-2.

[32] Michael L. Waskom. "Seaborn: Statistical Data Visualization". In: *Journal of Open Source Software* 6.60 (2021), p. 3021. DOI: 10.21105/joss.03021. URL: https://doi.org/10.21105/joss.03021.

[33] Zenda Woodman and Carolyn Williamson. "HIV Molecular Epidemiology: Transmission and Adaptation to Human Populations". In: *Current Opinion in HIV and AIDS* 4.4 (July 2009), pp. 247–252. ISSN: 1746-630X. DOI: 10.1097/COH.0b013e32832c0672. pmid: 19532060. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3351078/ (visited on 10/18/2023).